Exploratory Data Analysis on the Automobiles Dataset

Report

Introduction

The dataset is a .txt file which contains the information of many automobiles and their features and characteristics. Before cleaning, the file has 26 columns and 206 rows. Because the dataset has many columns, it is more efficient to place great focus on the data which will be used to generate our analysis and visualisations.

The data is a combination of many different variables which describe the vehicles with emphasis placed on the following:

- Make (Categorical - Nominal)
- Body Style (Categorical - Nominal)
- Wheel Base (Numerical - Continuous)
- Curb Weight (Numerical - Continuous)
- The length, width and height (Numerical - Continuous)
- Engine Size (Numerical - Continuous)
- Horsepower (Numerical - Continuous)
- City and Highway MPG (Numerical - Continuous)
- Price - (Numerical - Continuous)

Data Cleaning

When cleaning the data, there are a few things to look out for. To analyse accurately, the dataset must be complete and consistent. As per the guidelines, 2 columns are dropped as they are not used in the analysis. They are the "normalized-losses" and "symboling" columns. Then numerical data was rounded and converted to "int64". The name of one of the manufacturers was incorrectly recorded as “alfa romero” instead of “alfa romeo”. This was corrected. When checking for duplicate rows, none were found so none had to be removed.

Missing Data

The data initially had many "?" in random positions. This may be due to an error in data entry or the data was not available to be added. As a result these rows were removed leaving us with 193 rows when we removed rows with any instance of "?" as an input.

Data Stories and Visualisations

We are not told what the numbers in the dataset represent in terms of their values, but it can be assumed that a North American lens is to be applied to the data. This makes the most sense due to the values for prices and measurements. This would mean that any prices are measured in US Dollars, measurements are in inches etc.

Using the cleaned data, we can answer some questions and gain insights which help us answer subsequent questions through finding patterns and trends. The following questions will be answered:

- Which are the 5 most expensive cars?
- Which manufacturer builds the most fuel efficient vehicles?
- Which vehicles have the largest engine capacity?
- Which vehicle manufacturer has the most car models in the dataset

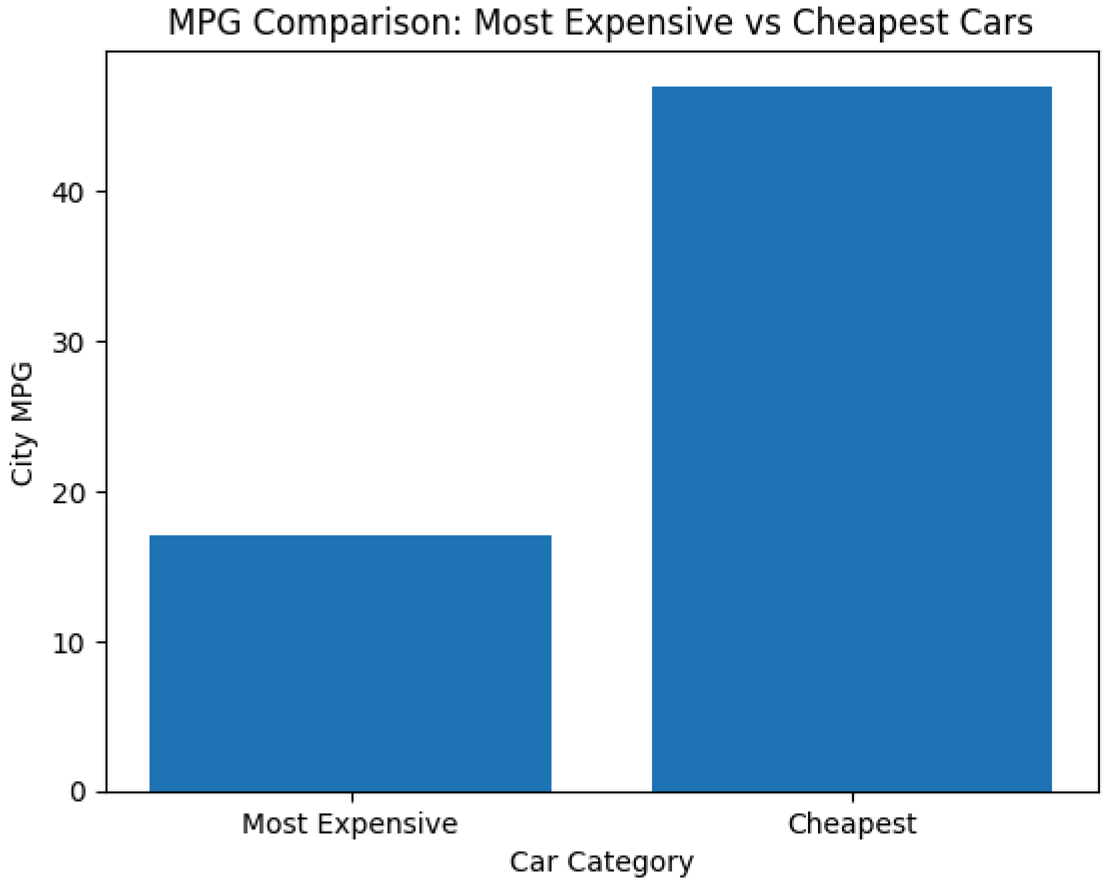
<u>Top 5 most expensive cars:</u>

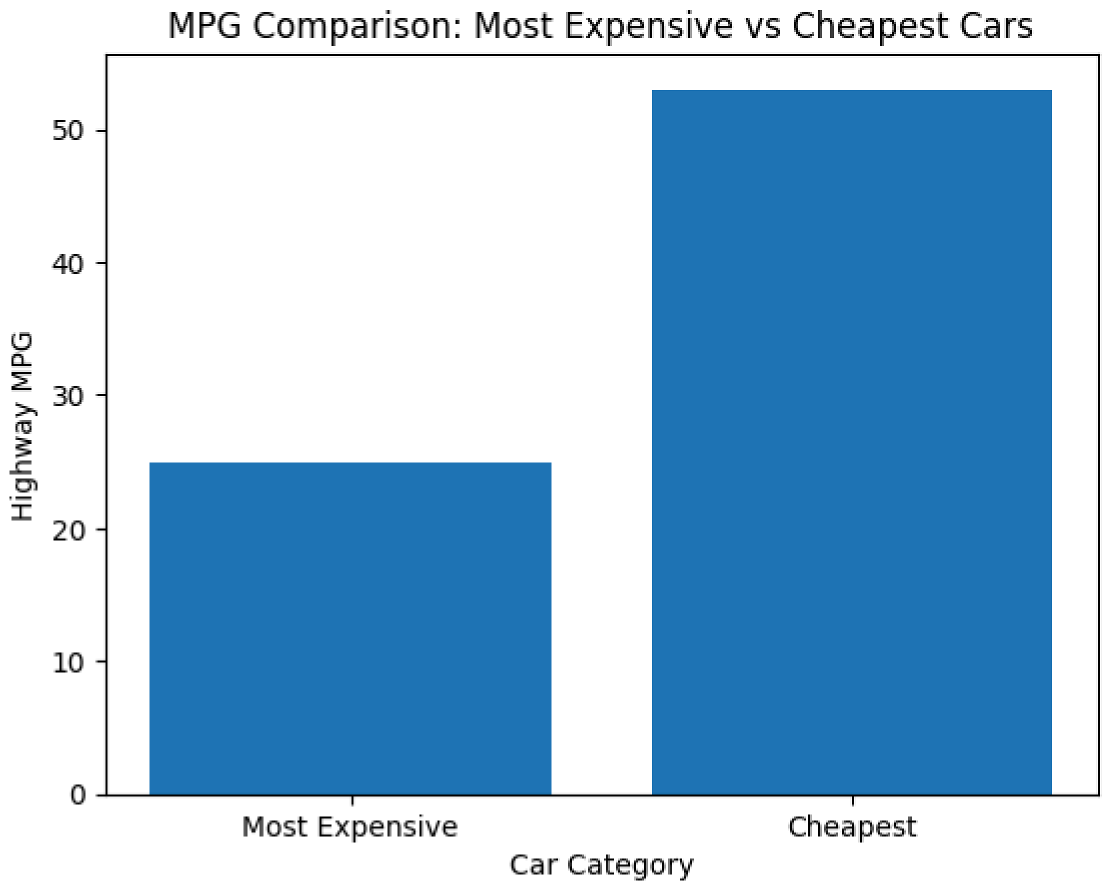
For the top 5 most expensive cars we have 2 from Mercedes-Benz, 2 from BMW and 1 from Porsche. These are European luxury brands so a more expensive price is to be expected. The dataset does not show us which models these cars are. This information may have been interesting and may have added some context to our findings, but we can look at other factors to help answer questions to follow. Below we can see the makes and prices of these cars:

```
Five Most Expensive Cars:

              make  price
74   mercedes-benz  45400
16             bmw  41315
73   mercedes-benz  40960
128        porsche  37028
17             bmw  36880
```

The numbers to the far left are simply the index of that car in the dataset. This will be present in subsequent instances where the information in the Jupyter notebook is shared within the report, but I just wanted to note what these numbers represent so it is clear.

When it comes to using MPG as a metric for efficiency, higher is better as you are covering more miles with each gallon of fuel consumed. We can also compare the city and highway MPG for both the most expensive and cheapest cars and look at fuel efficiency driving within the city and driving on the highway.


MPG Comparison: Most Expensive vs Cheapest Cars
City MPG
0
10
20
30
40
Most Expensive
Cheapest
Car Category


MPG Comparison: Most Expensive vs Cheapest Cars
Highway MPG
0
10
20
30
40
50
Most Expensive
Cheapest
Car Category

The bar charts show us that city MPG for the 5 cheapest cars is much better than the 5 most expensive cars. The same is true for highway MPG.

The physical dimensions, curb weights and body styles of both the most expensive and cheapest cars are shown below:

Most Expensive:

	make	curb-weight	length	width	height	body-style
74	mercedes-benz	3715	199	72	55	hardtop
16	bmw	3380	194	68	54	sedan
73	mercedes-benz	3900	208	72	57	sedan
128	porsche	2800	169	65	52	convertible
17	bmw	3505	197	71	56	sedan

Cheapest:

	make	curb-weight	length	width	height	body-style
138	subaru	2050	157	63	54	hatchback
18	chevrolet	1488	141	60	53	hatchback
50	mazda	1890	159	64	54	hatchback
150	toyota	1985	159	64	54	hatchback
76	mitsubishi	1918	157	64	51	hatchback

The data shows us a few possible reasons for this vast difference between City MPG values. The most expensive cars tend to have a bigger physical footprint (greater wheel base, length, width etc.) than the cheapest cars.

Similarly, we can look at the engine sizes, and horsepower numbers the most expensive and cheapest cars:

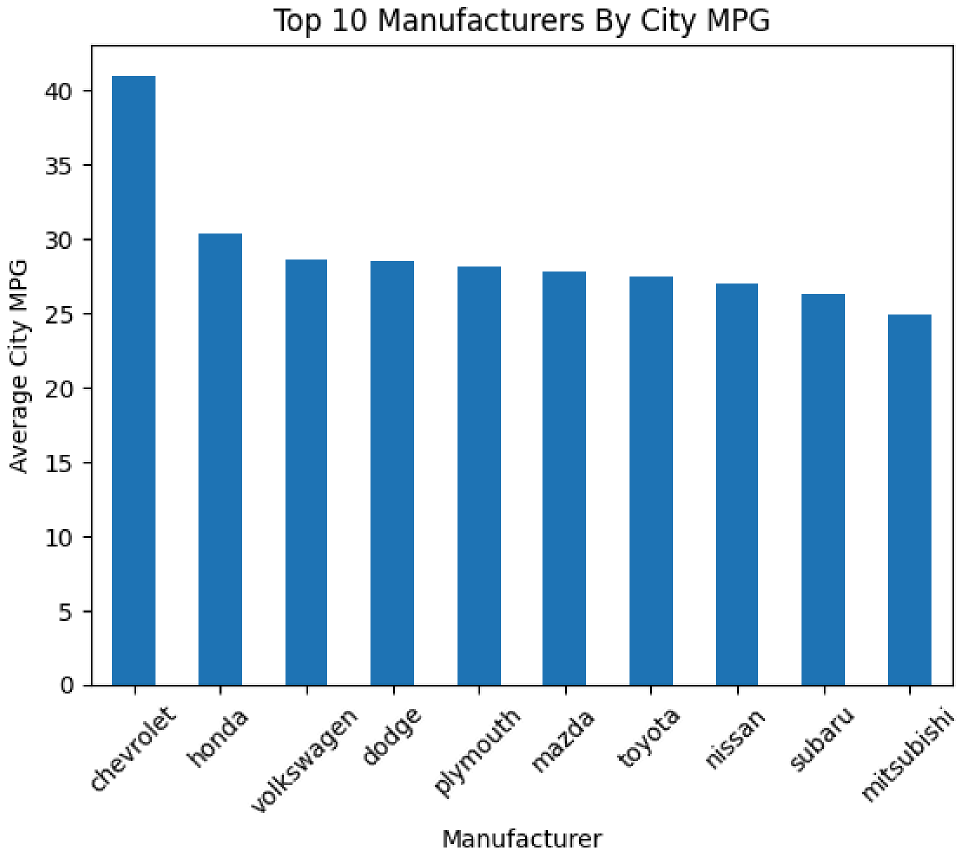
```
Most Expensive:

              make  engine-size  horsepower
74   mercedes-benz          304         184
16             bmw          209         182
73   mercedes-benz          308         184
128        porsche          194         207
17             bmw          209         182


Cheapest:

           make  engine-size  horsepower
138      subaru           97          69
18    chevrolet           61          48
50        mazda           91          68
150      toyota           92          62
76   mitsubishi           92          68
```

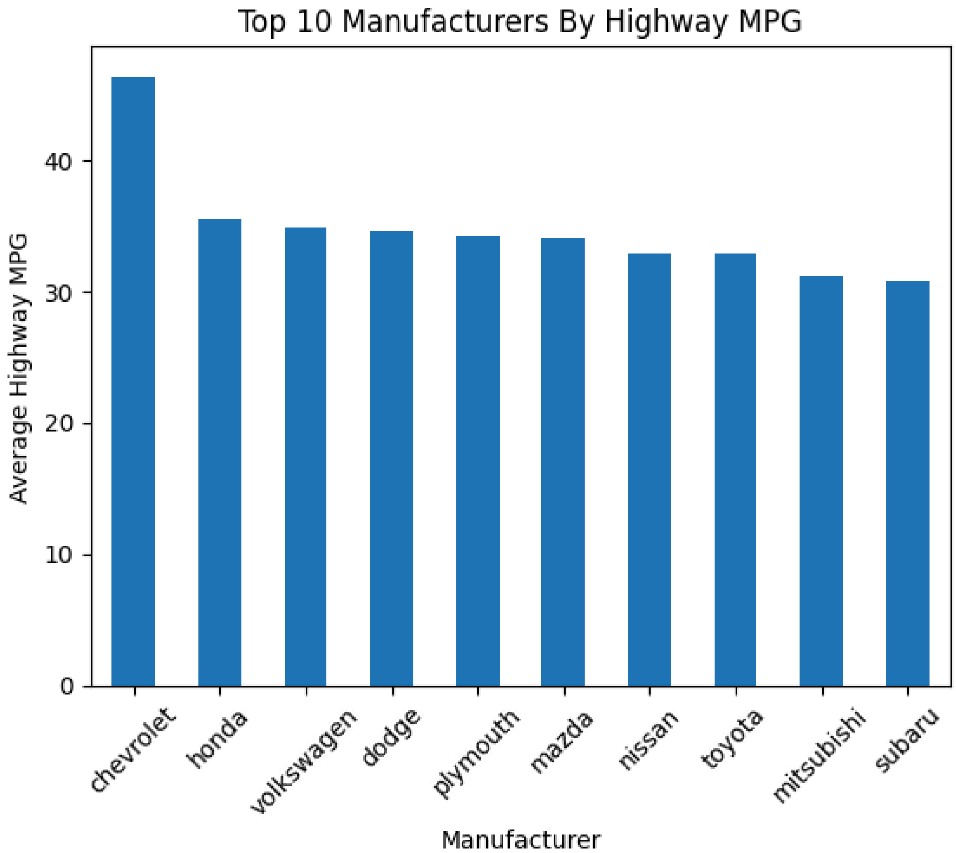
The most expensive cars are also more powerful cars with larger engines and greater curb weights than the cheapest cars. The cheapest cars are all hatchbacks, while the most expensive cars are sedans, a hardtop and a convertible.

The features of the most expensive and cheapest cars are not the sole reasons for why MPG is either bad or good, but are rather tools which can help demonstrate that there may be tendencies within the dataset which are present.

Cars that have greater horsepower tend to go through more fuel than cars with lower horsepower in order to achieve their higher horsepower numbers. A heavy car will require more fuel to create the power required to drive its mass than a lighter car would. A larger engine tends to create more power and weigh more than a smaller engine.

Which manufacturer builds the most fuel efficient cars?:

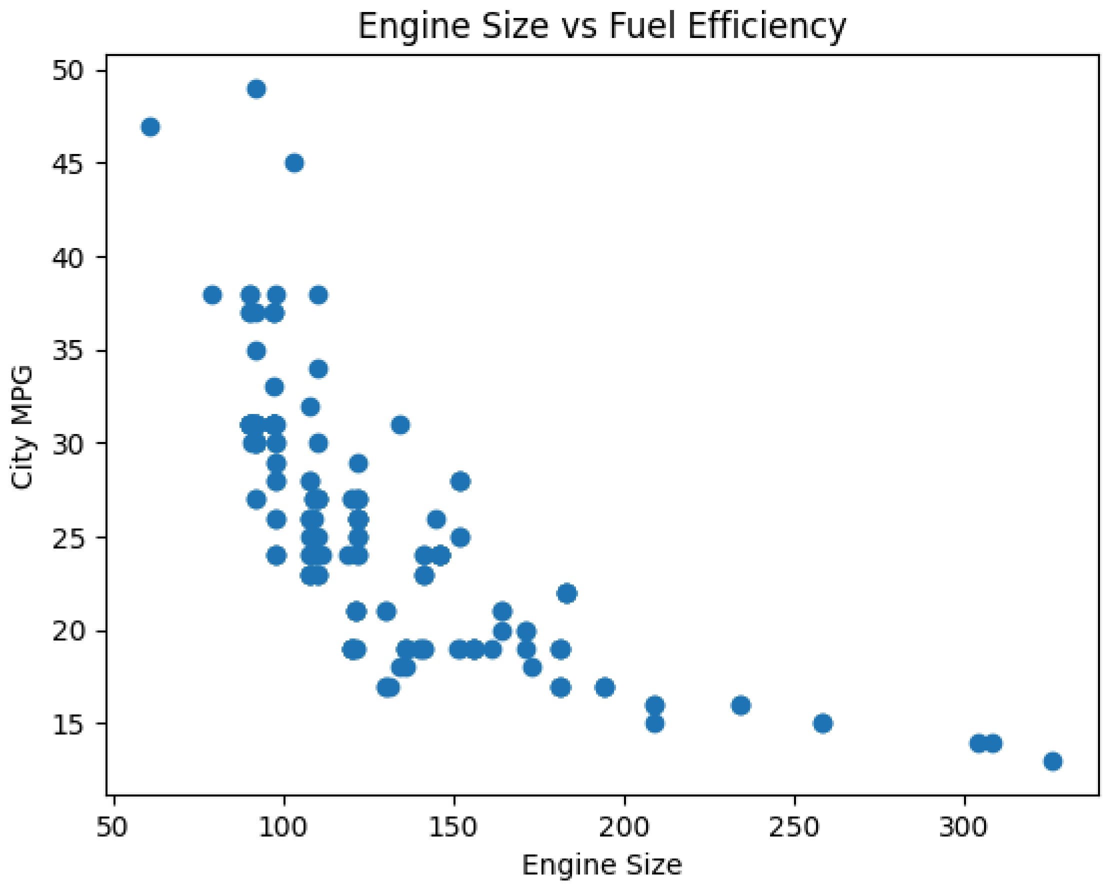




Chevrolet is the brand with the most fuel efficient cars. As we can see from the bar charts, they are clear victors when it comes to both city and highway MPG. When comparing average prices of the vehicles in the dataset, Chevrolet is the manufacturer with the cheapest average vehicle price. This supports what we know about the relationship between price and MPG.

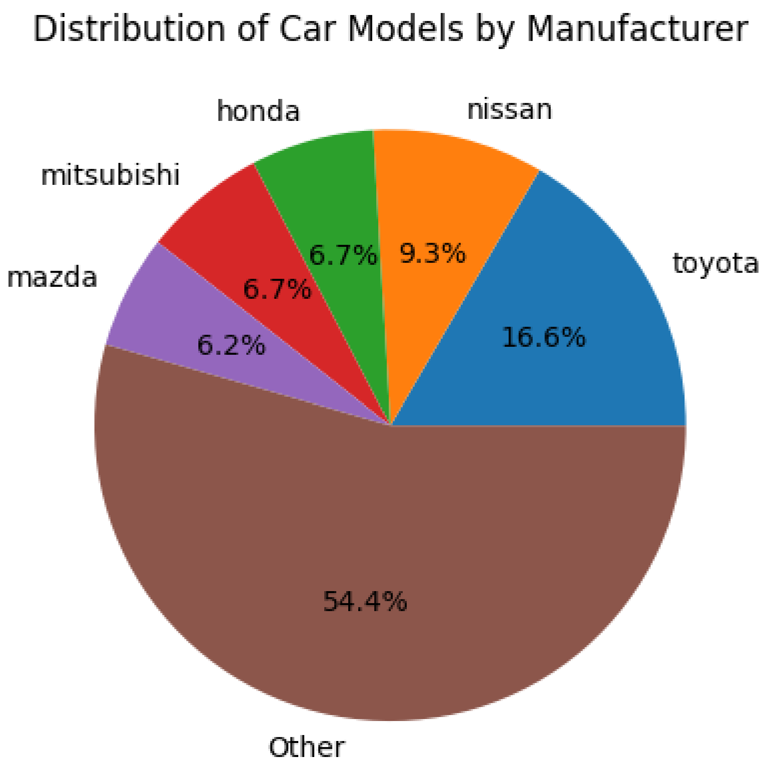
Which vehicles have the largest engines?:

The 5 vehicles with the largest engines are 3 from Jaguar and 2 from Mercedes-Benz. 4 of them are sedans and 1 is a hardtop.



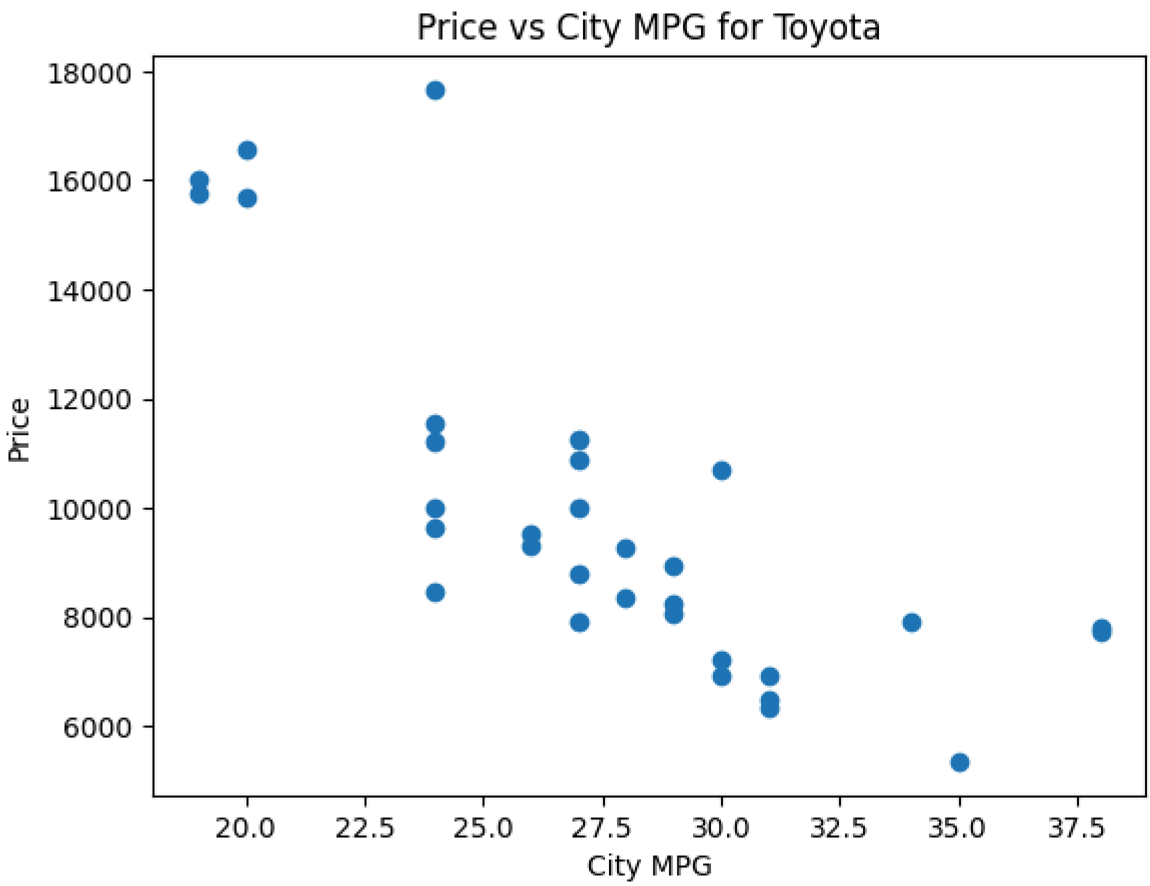
As seen in the scatter plot showing city MPG of all vehicles compared to their engine sizes, cars with larger engine sizes tend to be less fuel efficient and these 5 cars support this as they have very low city MPG (ranging from 13 to 15) and this trend is expressed well in the visualisation. There is a large cluster of more fuel efficient cars to the left where engine size is lower and on the right we see outliers but looking at the data, this is accurate.

Which manufacturer has the most car models in the dataset?:



The manufacturer with the most instances in the dataset is Toyota with 16.6% of the entries in the dataset. It is also very noteworthy that alongside Toyota, the other 4 major manufacturers in terms of representation in the data are also Japanese manufacturers (Mazda, Mitsubishi, Honda and Nissan).

For many, the purchase of a car is influenced by key factors such as cost and fuel efficiency. Most people just need a form of transport and are likely to make a buying decision based on what is best for their financial situation. These cars tend to be cheaper hatchbacks. As we have seen above, they are small and fuel efficient. They also tend to be used in the city most often.



As we can see from the scatter plot, the majority of the cars from Toyota are lower in price and have a decent city MPG. This is likely why they have such a good representation in the market for vehicles.

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